



University of St.Gallen

Institute of Computer Science



ETRA 2026

## EYE-CARE:

# Eye Tracking for Context-Adaptive Systems Based on Activity and Cognitive Load Prediction

June 1<sup>st</sup> - 4<sup>th</sup> 2026, Marrakech, Morocco

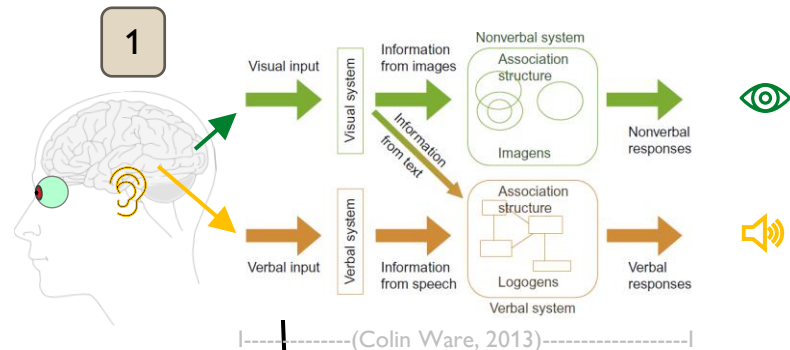
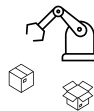
Kenan Bektaş and Amine Abbad-Andaloussi

From insight to impact.



## Kenan studies . . .

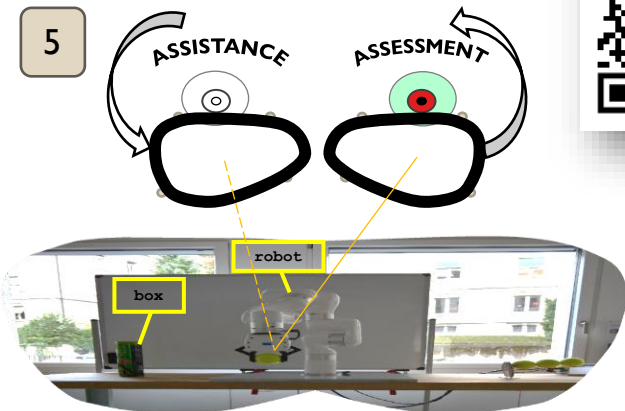
1. Dual Coding Theory ([Allan Paivio](#))
2. Cognitive Load (CL) Theory ([John Sweller](#))
3. Cognitive Offloading ([Risko and Gilbert](#))
4. Human Augmentation ([Raisamo et al.2019](#); [Villa et al., 2023](#))
5. Pervasive Eye Tracking ([Bulling and Gellersen, 2010](#); [Bektas, 2020](#))



- 2
- Intrinsic CL
  - Extraneous CL
  - Germane CL

3

**“Use physical action to alter the information processing requirements of a task to reduce cognitive demand”**



Motor-Sensory-Cognitive-Augmentation

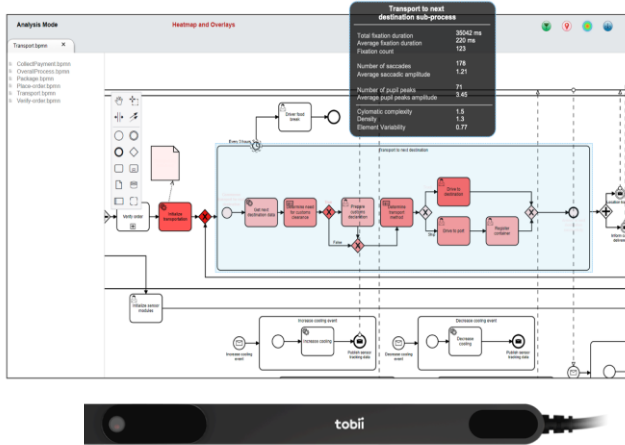


4

**“Technologies**  
that help users overcome limitations  
and integrate with their body”

**Toward A Pervasive Gaze-Contingent Assistance System:  
Attention and Context-Awareness in Augmented Reality**

# Eye-tracking Methods & Frameworks



## Adaptive Systems



1.) Biological Signal (e.g., EEG, pupil dilation, HRV, skin conductance) is recorded from the user

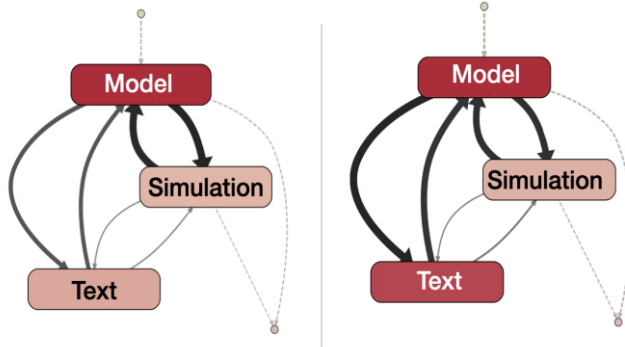


2.) Biological signal is analyzed to derive a cognitive/behavioral state



3.) System adapts on the base of the user's cognitive/behavioral state

## Cognitive & Behavioral Aspects



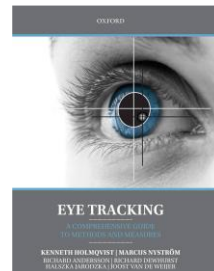
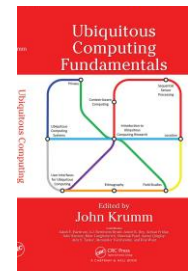
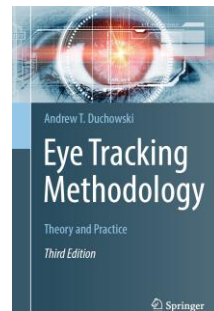
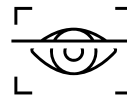
Learn more about my research in this stream

# Ubiquitous Computing and Context-Aware Computing



Seamless integration of networked (micro-) computers and displays into the physical world.

Systems that can adopt to the **User**, its **Physical-** and **Computational-**environment



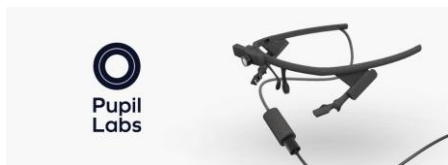
# Put It There



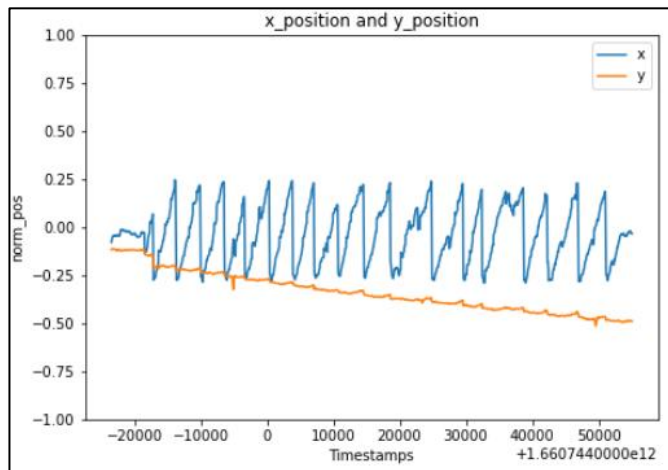
Richard Bolt and colleagues in 80's (MIT)

| Computer Graphics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Volume 15, Number 3 | August 1981 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------|
| GAZE-ORCHESTRATED DYNAMIC WINDOWS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                     |             |
| Richard A. Bolt<br>Architecture Machine Group<br>Massachusetts Institute of Technology<br>Cambridge, Massachusetts 02139                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                     |             |
| ABSTRACT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                     |             |
| <p>Consider a large-format display before the user, bearing a multiplicity of "windows," like little movies, the majority dynamic and in color. There are upwards of 20 windows, say, more than a person can ordinarily absorb at once. Some of the windows come and go, reflecting their nature as direct TV linkages into real-time, real-world events. Others are non-real-time, some dynamic, others static but capable of jumping into motion.</p> <p>Such an ensemble of information inputs reflects the managerial world of the top-level executive of the not too distant electronic future: a world of brevity, fragmentation, variety, above all one of an overwhelming onslaught of events.</p> <p>The multiplicity and simultaneity of such a display situation ordinarily would make coping with it untenable. The intent of the reported research is to introduce order and control through the creation of a dynamic gaze-interacting interface.</p> <p>Making the behavior and reactivity of the "windows" contingent upon measured eye-movements - the point-of-regard of the observer - aims both to help the observer to cope with the onslaught of events on the one hand, yet enable on the other hand continuing close contact with that everchanging ensemble.</p> <p>A simulation of such a world is described and demonstrated in the composite medium of computer, videodisc, and video special effects. Eye-tracking technology, integrated with speech and manual inputs, controls the display's visual dynamics, and orchestrates its sound accompaniments. All elements are combined to form a testbed for the conception generally, and to explore the associated human factors and stagecraft.</p> |                     |             |
| Key Words: Eye-tracking; videodisc; man-machine interfaces; graphical interface                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                     |             |
| The work reported herein has been supported by the Cybernetics Technology Division of the Defense Advanced Research Projects Agency, under Contract No. MDA-903-77-C-0037.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                     |             |
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| ©1981 ACM 0-8971-045-1/81/0805-0309 \$00.75                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                     |             |

# What is Eye Tracking Data?



Eye  
tracker



## Time series data

|    | eyeDataTimestamp | gazeHasValue | gazeOrigin_x | gazeOrigin_y | gazeOrigin_x |
|----|------------------|--------------|--------------|--------------|--------------|
| 1  | 1660743976560    | True         | 0.00688      | 0.06962      | 0.00688      |
| 2  | 1660743976594    | True         | 0.00616      | 0.06952      | 0.00616      |
| 3  | 1660743976627    | True         | 0.00529      | 0.06944      | 0.00529      |
| 4  | 1660743976660    | True         | 0.00506      | 0.06944      | 0.00506      |
| 5  | 1660743976694    | True         | 0.00494      | 0.06947      | 0.00494      |
| 6  | 1660743976727    | True         | 0.00470      | 0.06947      | 0.00470      |
| 7  | 1660743976760    | True         | 0.00399      | 0.06940      | 0.00399      |
| 8  | 1660743976794    | True         | 0.00281      | 0.06946      | 0.00281      |
| 9  | 1660743976827    | True         | 0.00191      | 0.06958      | 0.00191      |
| 10 | 1660743976860    | True         | 0.00132      | 0.06972      | 0.00132      |

# Activity Prediction from Eye Movements

## 1. Data Collection

- Understand the activity: For example, *Reading* material (paper or digital etc.), geometric constraints, static vs. dynamic scene (scrolling) etc.
- Understand the temporal aspects that are relevant for data collection and model development (seconds to hours)
- Collect ground truth data

## 2. Model Development

- Calculate eye movement metrics (fixation, saccade, blinks, etc.)
- Data Preprocessing (cleaning, smoothing, interpolation, etc.)
- Calculate features (normalization)
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- Train models and document their performance

## 3. Feedback

- What do we do with the detected activity?



# Notebook 1:

|                                           |                                         |
|-------------------------------------------|-----------------------------------------|
| 📁 Data/RawGazeData                        | Delete Data/RawGazeData/placeholder.txt |
| 📁 demo_images                             | intervention example added              |
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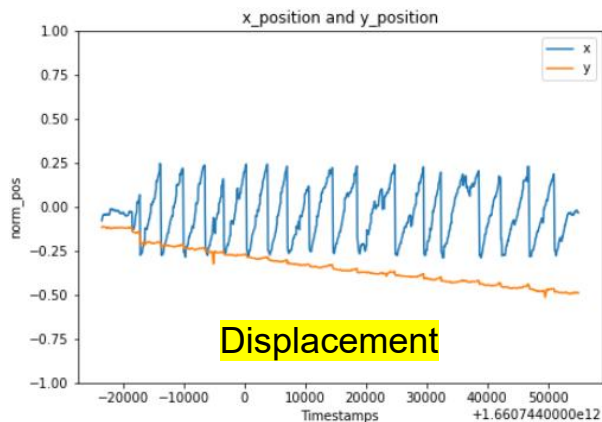


# Eye Movement Measures



## Time Series

|    | eyeDataTimestamp | gazeHasValue | gazeOrigin_x | gazeOrigin_y | gazeOrigin_x |
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Salvucci and Goldberg 2000

I-DT (protocol, dispersion threshold, duration threshold)

While there are still points

Initialize window over first points to cover the duration threshold

If dispersion of window points  $\leq$  threshold

Add additional points to the window until dispersion  $>$  threshold

Note a fixation at the centroid of the window points

Remove window points from points

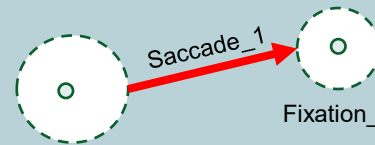
Else

Remove first point from points

Return fixations



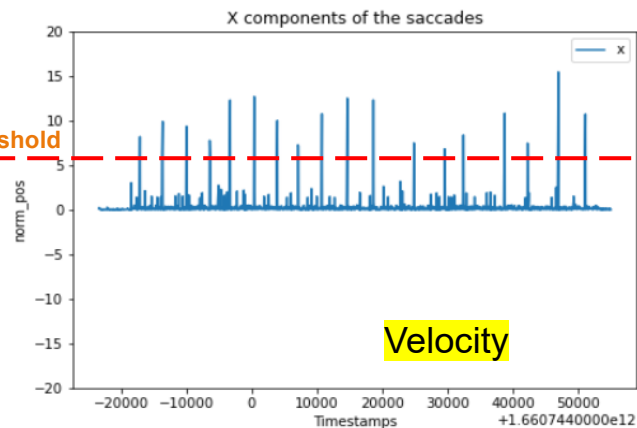
Dispersion size and duration



Fixation\_1

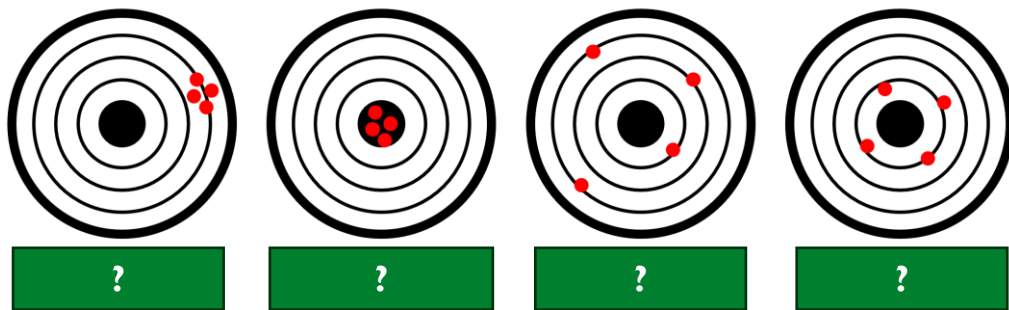
Fixation\_2

Velocity Threshold



# Eye Tracking Data

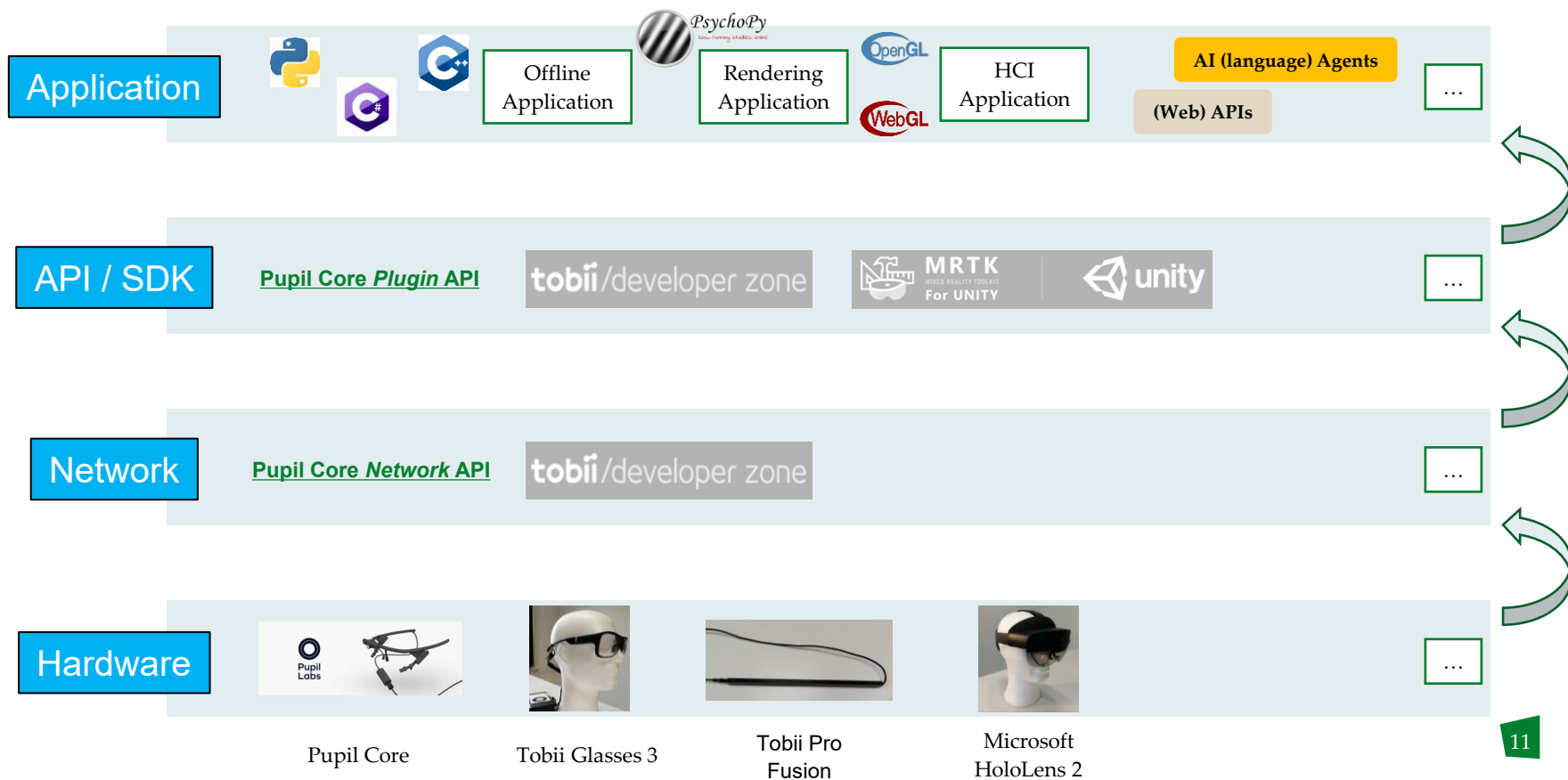
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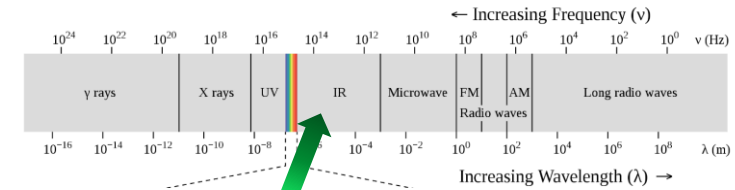
Kasneji et al 2024 <https://arxiv.org/abs/2404.15435>

- **Sampling rate**
  - *How many raw data per second?*
  - Measured in Hertz (Hz)
- **Accuracy**
  - *Deviation between the actual and measured*
  - Measured in degrees °
  - Depends on calibration quality and hardware
- **Precision**
  - *Deviation of the measurements*
  - Measured by the root mean square (RMS) error
  - Mainly hardware dependent
- **Calibration**
  - *A mapping between the measured and actual points of interest*
  - Pay attention to *slippage* (compensation) and *lighting* conditions

# Hardware and Software



# Eye Trackers (and MR-headsets)



## Mixed Reality (MR) Headsets

## Eye Tracking (ET) Devices

- Mobility
- Weight or Size
- Field-of-view
- Rendering quality
- Display resolution
  - Frame rate
- Connectivity
- Battery life
- App development
  - .....

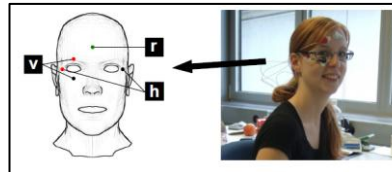
| Connectivity     | Wired                                                                             | Mobile or Wireless                                                                |                                                                                     | Wired                                                                               |
|------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Examples         | HTC VIVE Pro Eye,<br>Varjo XR3                                                    | Microsoft<br>HoloLens 2                                                           | Tobii Glasses 3,<br>Pupil Core                                                      | Tobii Pro Fusion,<br>EyeLink 1000 Plus                                              |
|                  |  |  |  |  |
| ET Sampling Rate | 120-200Hz                                                                         | 30Hz                                                                              | 50-200Hz                                                                            | 60-2000Hz                                                                           |
| ET Accuracy      | $\approx 1.1^\circ$                                                               | $\approx 1.5^\circ - 3.0^\circ$                                                   | $\approx 0.6^\circ - 0.8^\circ$                                                     | $\approx 0.2^\circ - 0.5^\circ$                                                     |

Head-Mounted

Remote  
or  
Desktop-Mounted

- Mobility
- Weight or Size
- Field-of-view
- Data quality or accuracy
  - Sampling rate
- Display resolution
  - Connectivity
- Battery life
- Open-source?
  - .....

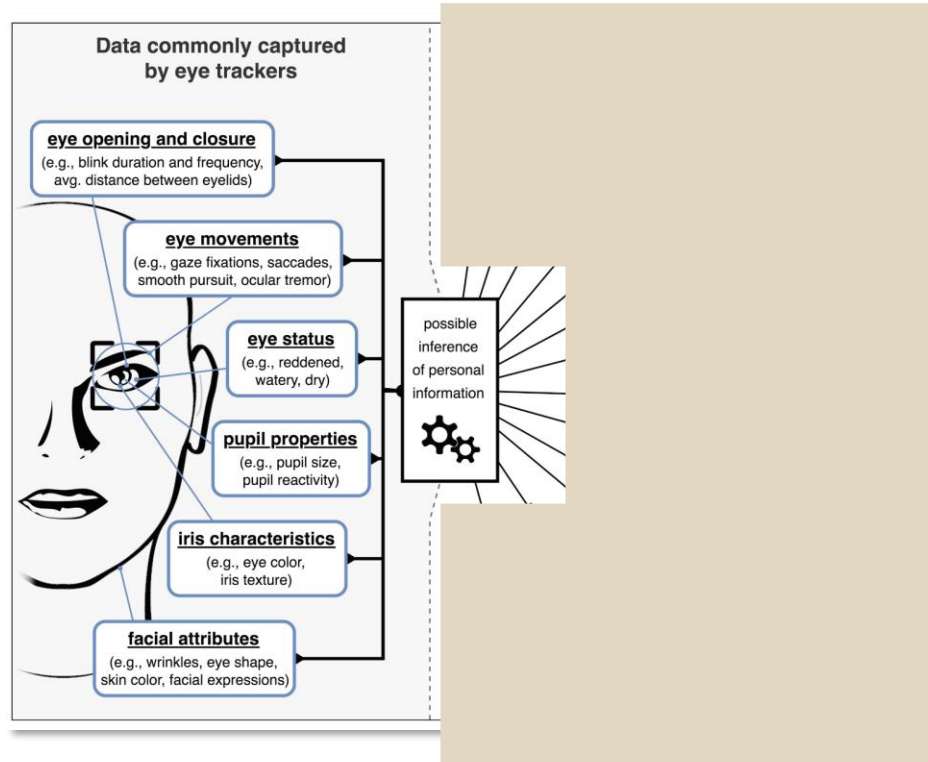
Bektaş, K., Strecker, J., Mayer, S., & Stolze, M. (2022). [EToS-1: Eye Tracking on Shopfloors for User Engagement with Automation](#). In *AutomationXP22: Engaging with Automation, CHI'22*



Eye Movement Analysis for Activity Recognition  
Using Electrooculography

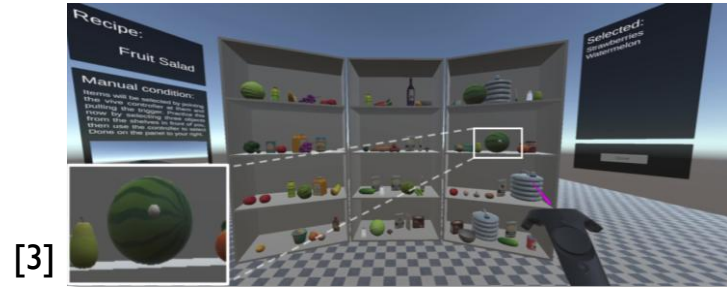
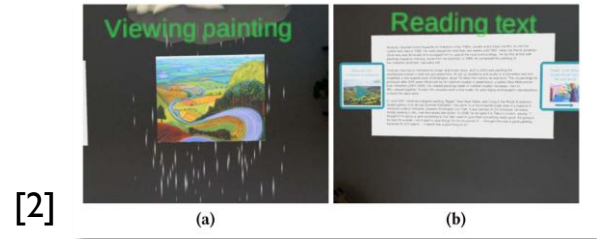
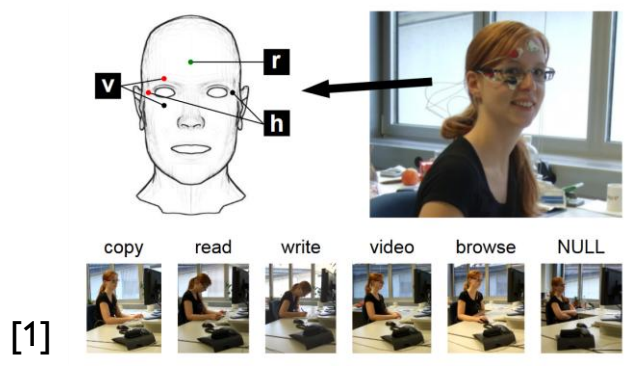
Andreas Bulling, Student Member, IEEE, Jamie A. Ward,  
Hans Gellersen, and Gerhard Tröster, Senior Member, IEEE

# What can **You** infer from Eye Tracking Data?



# Keep an **Eye** on your **Activities**

Eye Tracking can also be used to infer someone's activity!



- [1] Bulling et al. 2011. Eye Movement Analysis for Activity Recognition Using Electrooculography. IEEE TPAMI 33, 4 (April 2011), 741–753. <https://doi.org/10.1109/TPAMI.2010.86>
- [2] Scargill et al. 2022. Demo Abstract: Catch My Eye: Gaze-Based Activity Recognition in an Augmented Reality Art Gallery. IPSN'22. <https://doi.org/10.1109/IPSN54338.2022.00052>
- [3] David-John et al. 2021. Towards gaze-based prediction of the intent to interact in virtual reality. ETRA'21. <https://doi.org/10.1145/3448018.3458008>

# Keep an Eye on our Activities

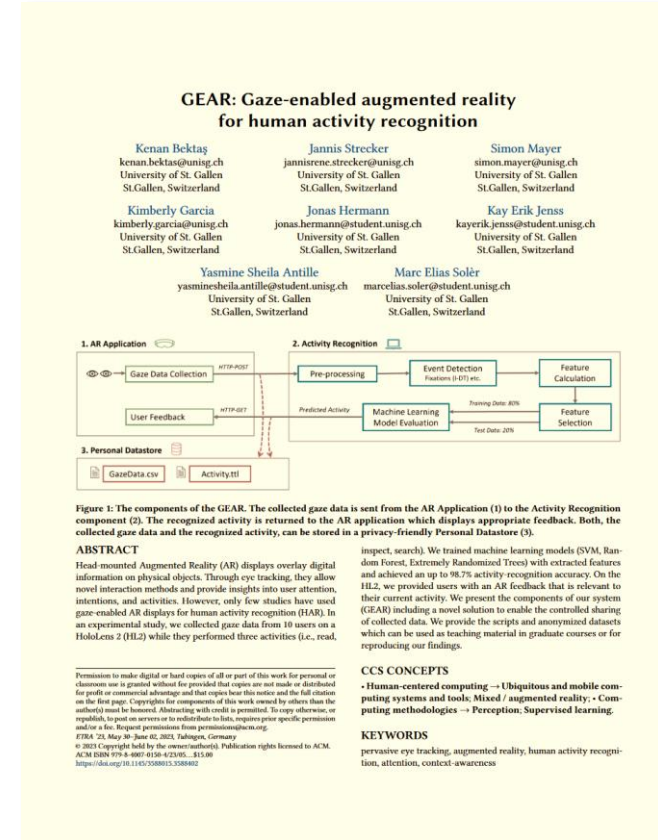
Output from three years ago:

## 1st Activity: Reading

Link to video: <https://www.youtube.com/watch?v=Dq-Z5p61J8E>

Kenan Bektaş, Jannis Strecker, Simon Mayer, Kimberly Garcia, Jonas Hermann, Kay Erik Jenß, Yasmine Sheila Antille, and Marc Solèr. 2023. GEAR: Gaze-enabled augmented reality for human activity recognition. In Proceedings of the 2023 Symposium on Eye Tracking Research and Applications (ETRA '23).

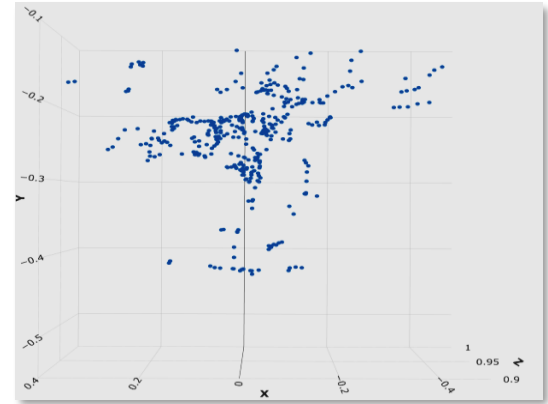
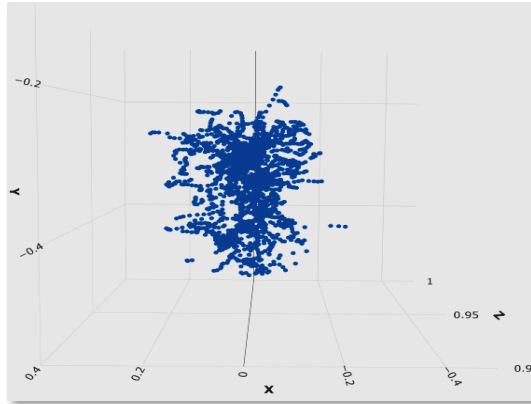
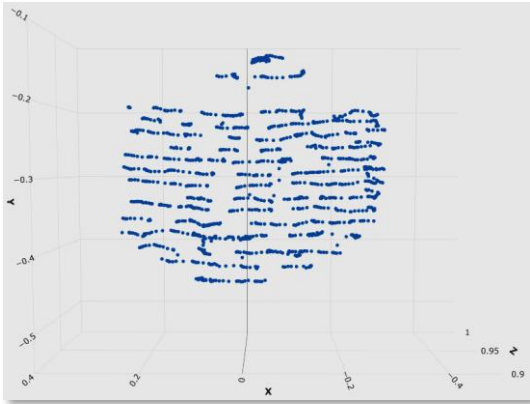
<https://doi.org/10.1145/3588015.3588402>





# Task 1: Offline Gaze Data Analysis for Activity Recognition

Which gaze data belongs to which activity?

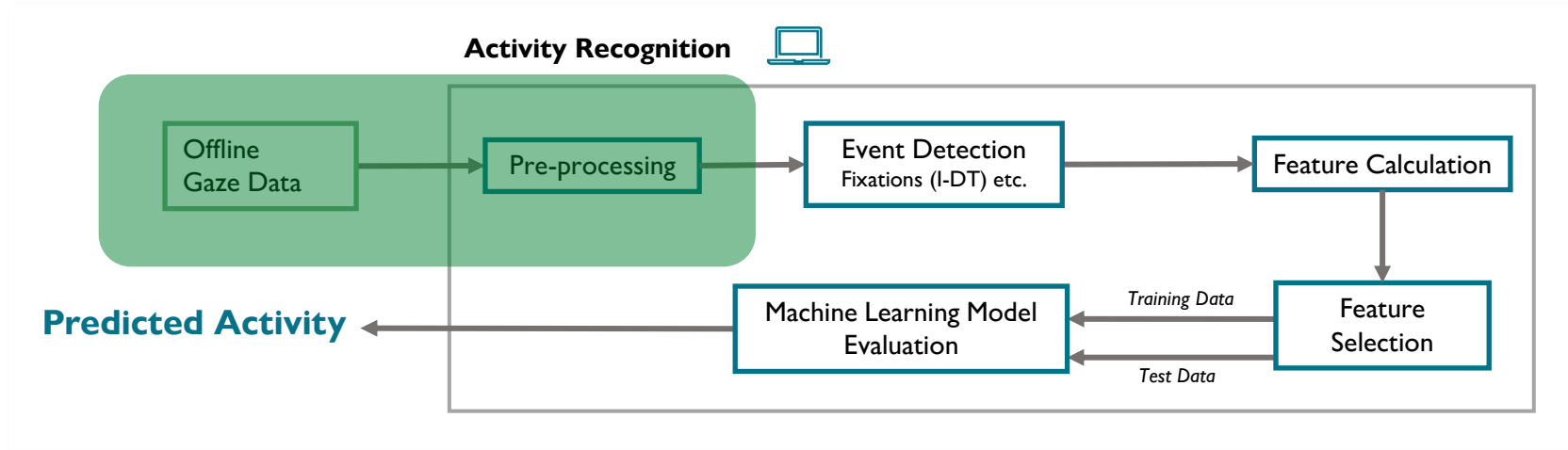


# Offline Gaze Data Analysis for Activity Recognition



Develop a model that classifies given eye movement data to estimate the current activity of a user.

Pipeline:



# From Eye Movement Metrics to Features

| Type     | Duration (ms) | Amplitude | Velocity | Example                                                  |
|----------|---------------|-----------|----------|----------------------------------------------------------|
| Fixation | 200-300       | -         | -        | When the eye temporarily stops at a word during reading. |
| Saccade  | 30-80         | 4-20°     | 30-500°  | When the eye moves from one object to another.           |
| (Blink)  | -             | -         | -        | When the eye is closed for a short moment of time.       |

## Eye movement features



Descriptive statistics of the eye movement metrics e.g.,

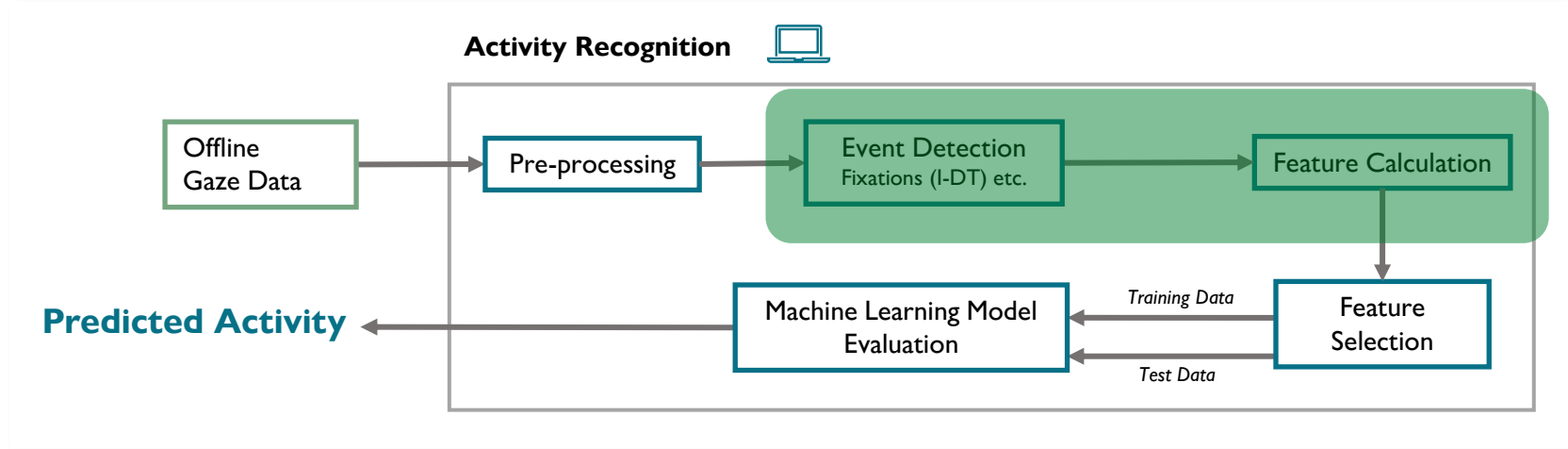
- (min/max/avg) fixation duration
- fixation dispersion
- fixation density
- blink rate per minute

# Offline Gaze Data Analysis for Activity Recognition



Develop a model that classifies given eye movement data to estimate the current activity of a user.

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# Activity Prediction from Eye Movements

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## Notebook 2:



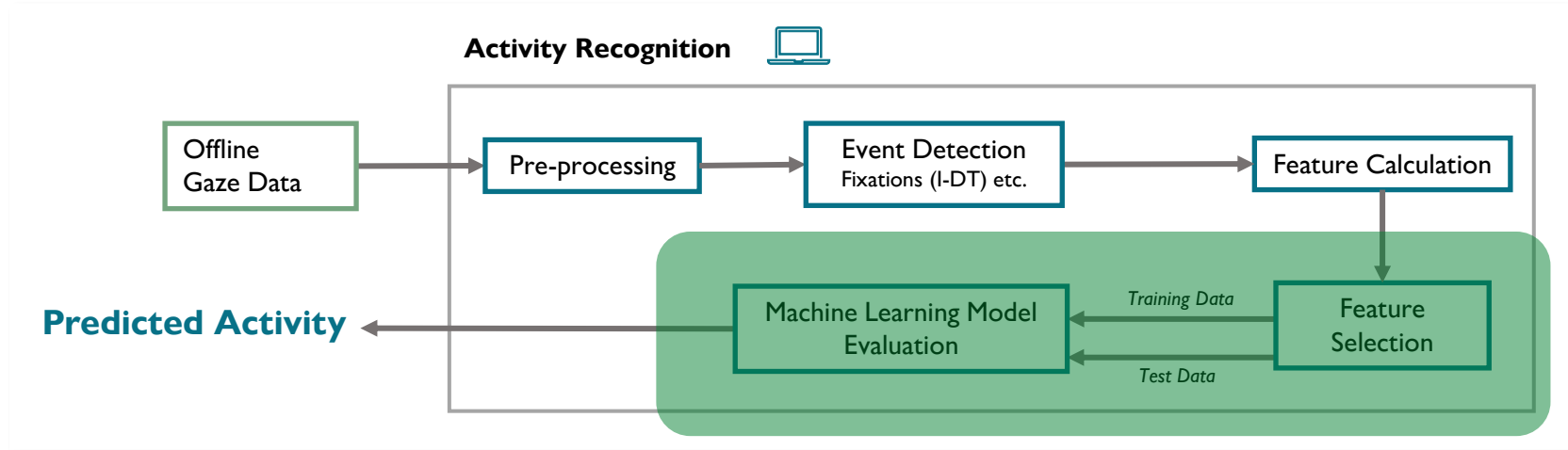
|                                           |                                         |
|-------------------------------------------|-----------------------------------------|
| 📁 Data/RawGazeData                        | Delete Data/RawGazeData/placeholder.txt |
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## Notebook 3:

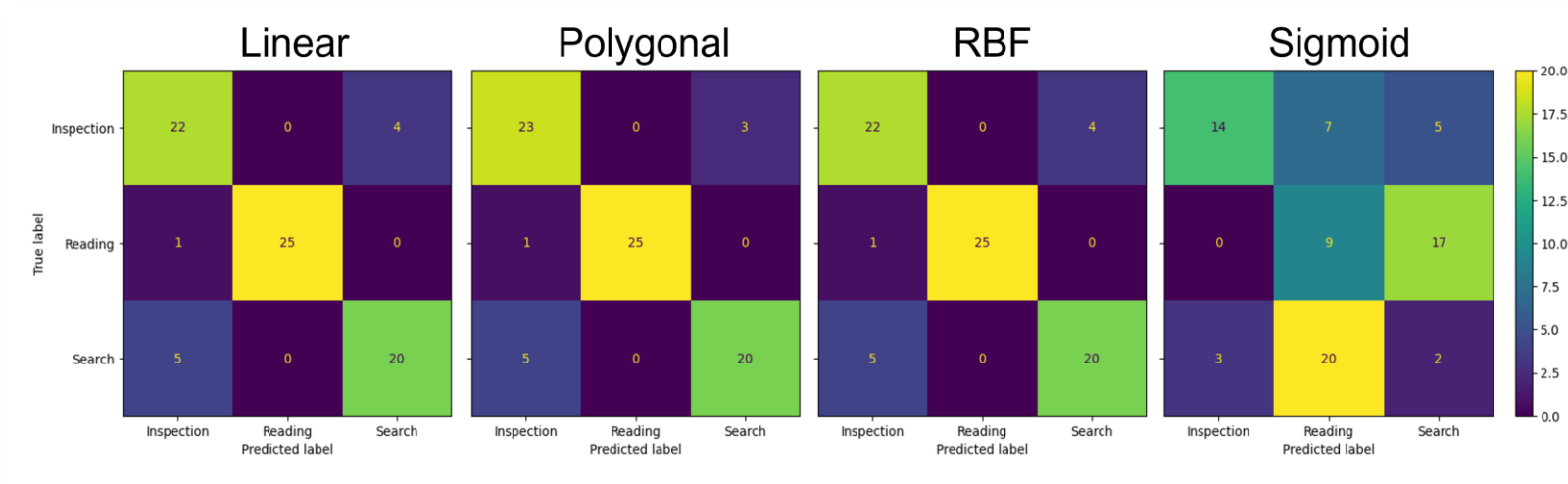


|                                           |                                         |
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# The existing SVM-based classifier

## Confusion Matrices:



## Accuracies:

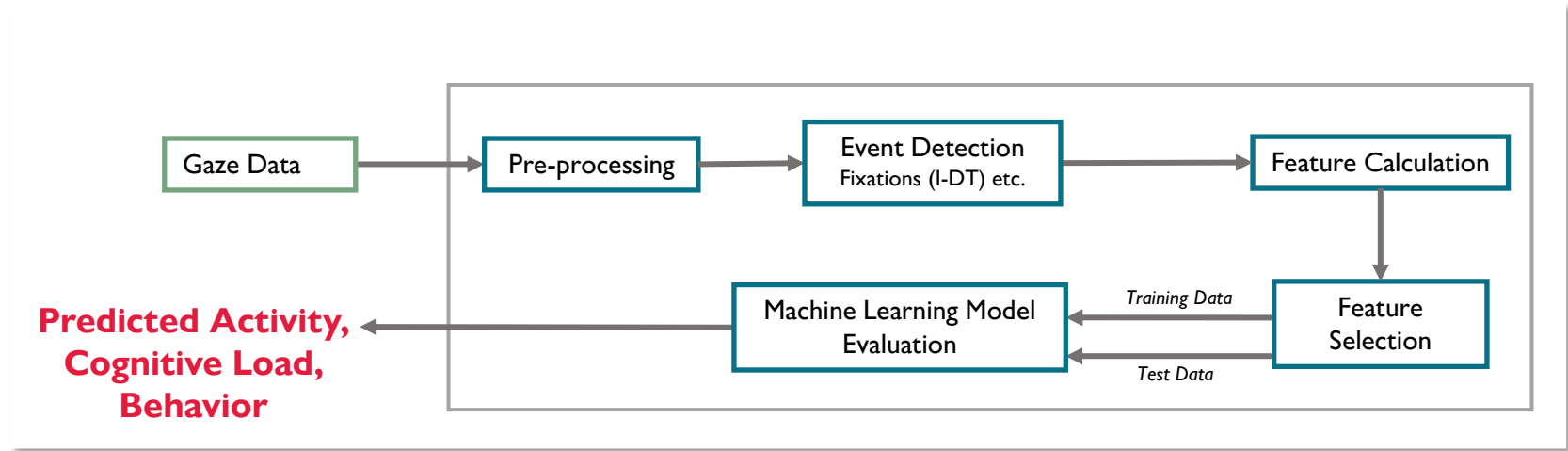
87.0%

88.3%

87.0%

32.4%

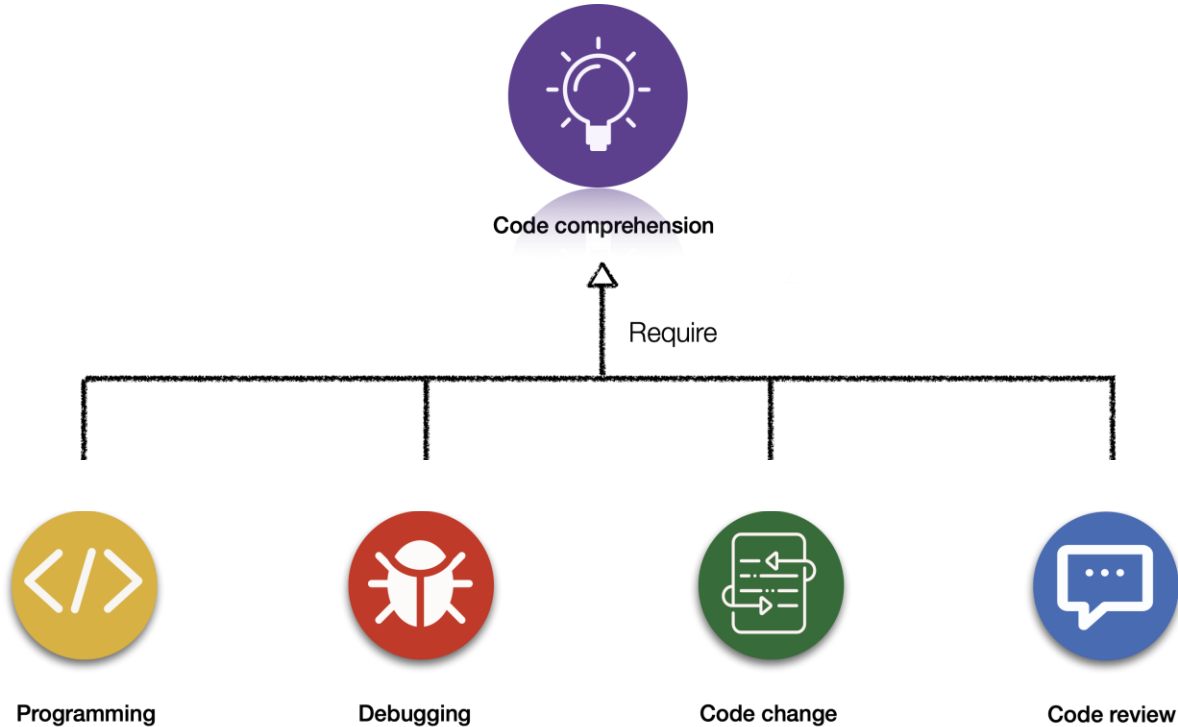
# Generalizing Beyond Activity Recognition



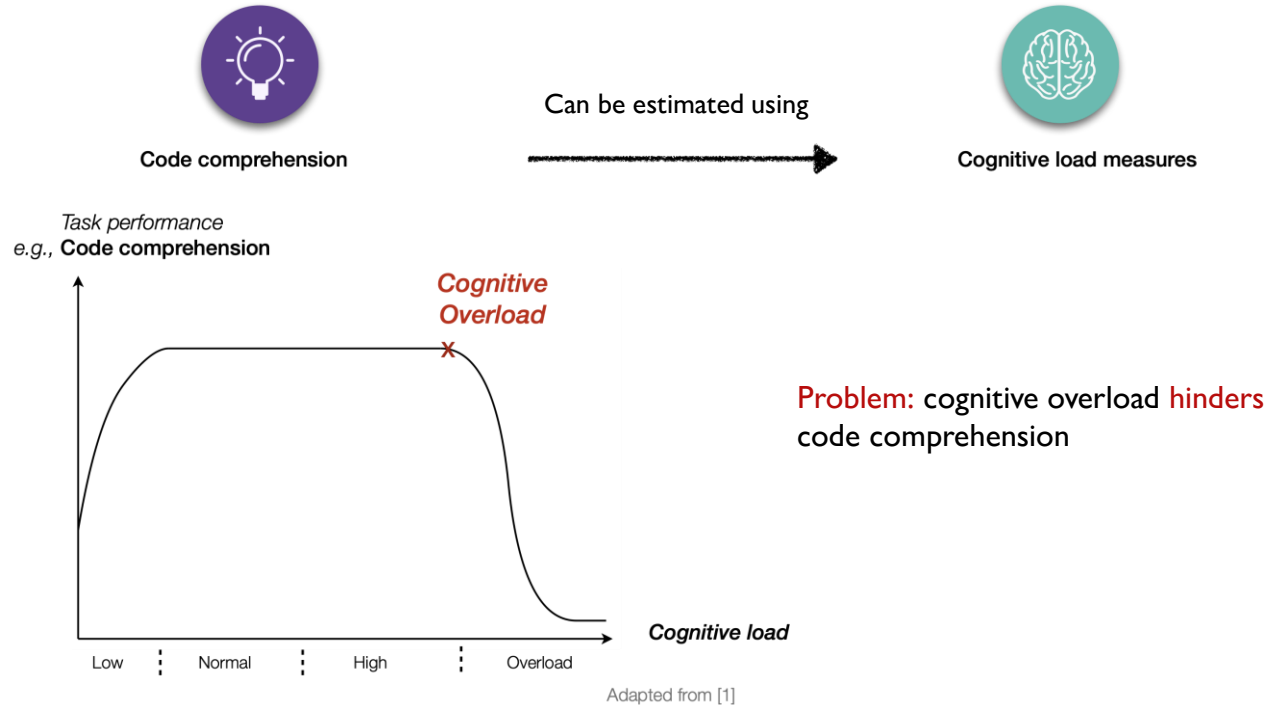
Similar pipeline. Different data; Different output

**Another case from the literature:** Cognitive Load Detection During Program Comprehension

# The Case of Cognitive Load Detection During Program Comprehension: Context



# The Case of Cognitive Load Detection During Program Comprehension: Context



# Cognitive Load Measures



Eye-tracking – Light weight, continuous measure

```
8= public static void main(String[] args) {
9     List<Object> array = new ArrayList<Object>();
10
11     Object r = new Rectangle();
12     array.add(r);
13     Object e = new Triangle();
14     Object s = new Circle();
15     array.add(s);
16     array.add(e);
17
18     for(int i=0; i<array.size(); i = next(i,array)){
19         Graphics.draw(array.get(i));
20     }
21 }
22
23= public static int next(int i, List<Object> array) {
24     if(array.get(i) instanceof Triangle) return array.size();
25     else if(array.get(i) instanceof Rectangle) return i+2;
26     return i-1;
27 }
28
```

A diagram illustrating eye-tracking data on the provided code. Green circles represent fixations, and arrows represent saccades. Fixations are located on the following lines: 11 (Object r), 14 (Object s), 19 (Graphics.draw), 24 (if statement), 25 (else if statement), and 26 (return statement). Saccades connect these fixations in a sequence: from line 11 to 14, 14 to 19, 19 to 24, 24 to 25, 25 to 26, and 26 back to 24.

Pupil measures

Fixation measures

Saccade measures

Scan-path measures

# Typical Black-Box Approaches

```
8= public static void main(String[] args) {
9     List<Object> array = new ArrayList<Object>();
10
11     Object r = new Rectangle();
12     array.add(r);
13     Object e = new Triangle();
14     Object s = new Circle();
15     array.add(s);
16     array.add(e);
17
18     for(int i=0; i<array.size(); i = next(i,array)){
19         Graphics.draw(array.get(i));
20     }
21 }
22
23= public static int next(int i, List<Object> array) {
24     if(array.get(i) instanceof Triangle) return array.size();
25     else if(array.get(i) instanceof Rectangle) return i+2;
26     return i-1;
27 }
28
```



Estimate cognitive load at the level of the **entire source-code**

e.g., (Fritz et al. 2014)

## Challenges:

- Not all the parts of the code have the same level of difficulty
- One **cannot pinpoint** the mentally demanding parts of code with the black-box approaches

# A White-Box Approach

```
8= public static void main(String[] args) {
9     List<Object> array = new ArrayList<Object>();
10
11     Object r = new Rectangle();
12     array.add(r);
13     Object e = new Triangle();
14     Object s = new Circle();
15     array.add(s);
16     array.add(e);
17
18     for(int i=0; i<array.size(); i = next(i,array)){
19         Graphics.draw(array.get(i));
20     }
21 }
22
23= public static int next(int i, List<Object> array) {
24     if(array.get(i) instanceof Triangle) return array.size();
25     else if(array.get(i) instanceof Rectangle) return i+2;
26     return i-1;
27 }
28
```



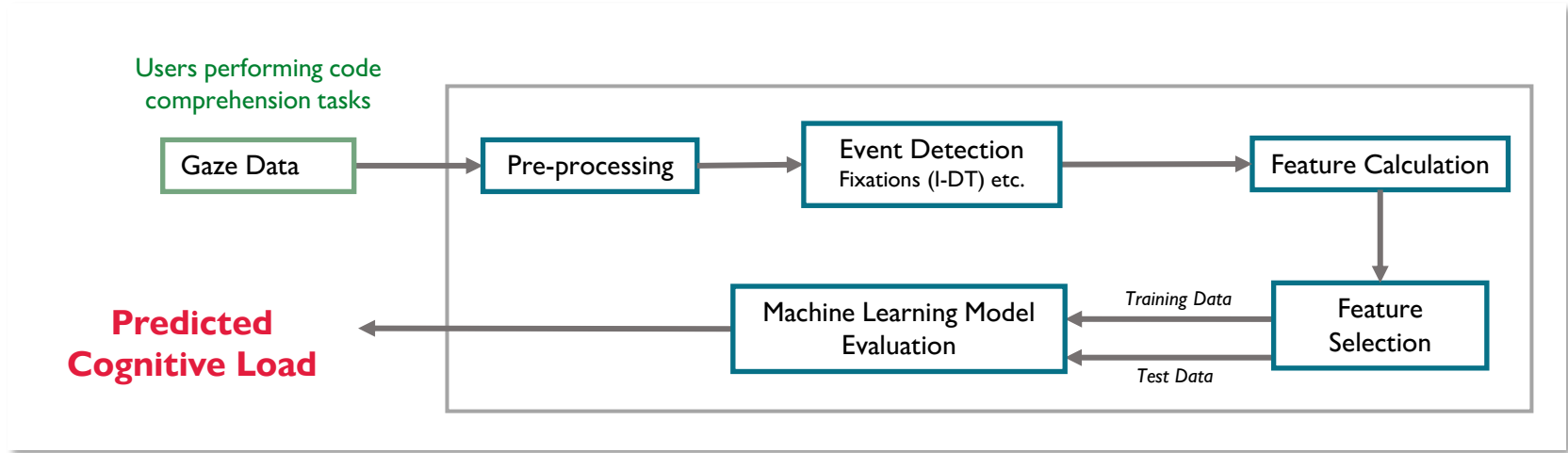
Estimate cognitive load at a fine-grained level e.g., each line of code

e.g., (Abbad-Andaloussi et al. 2022)

## Benefits:

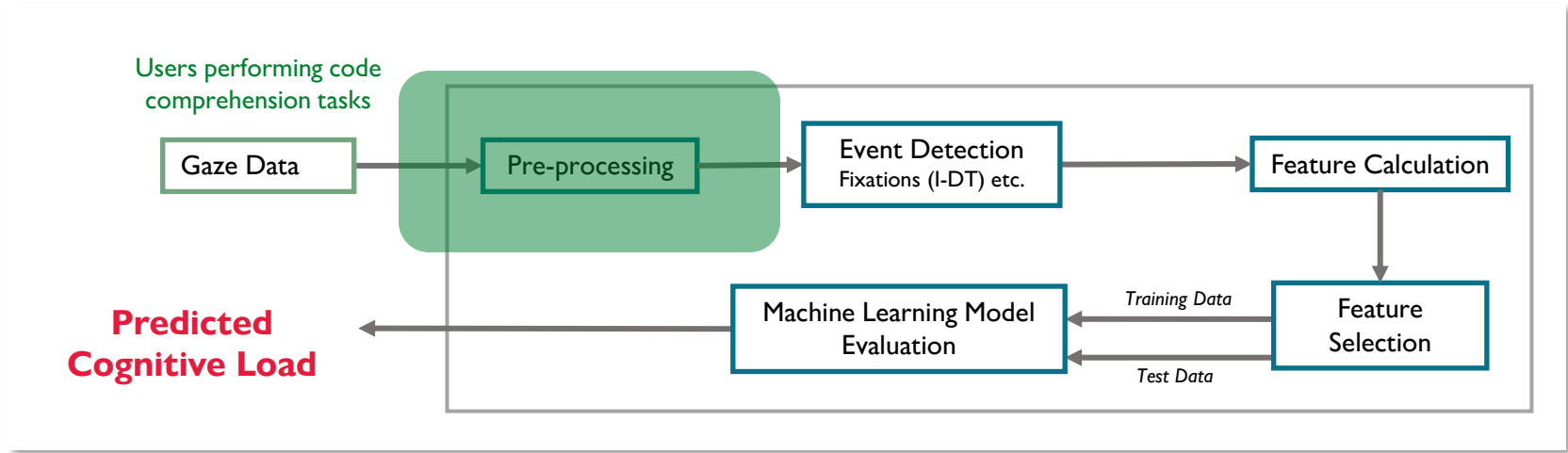
- Pinpoint the exact parts that are mentally demanding in the source-code
- Subsequently, focus on the mentally demanding parts of code during refactoring and code review tasks to enhance the comprehensibility of the source-code

# Back to Our Pipeline





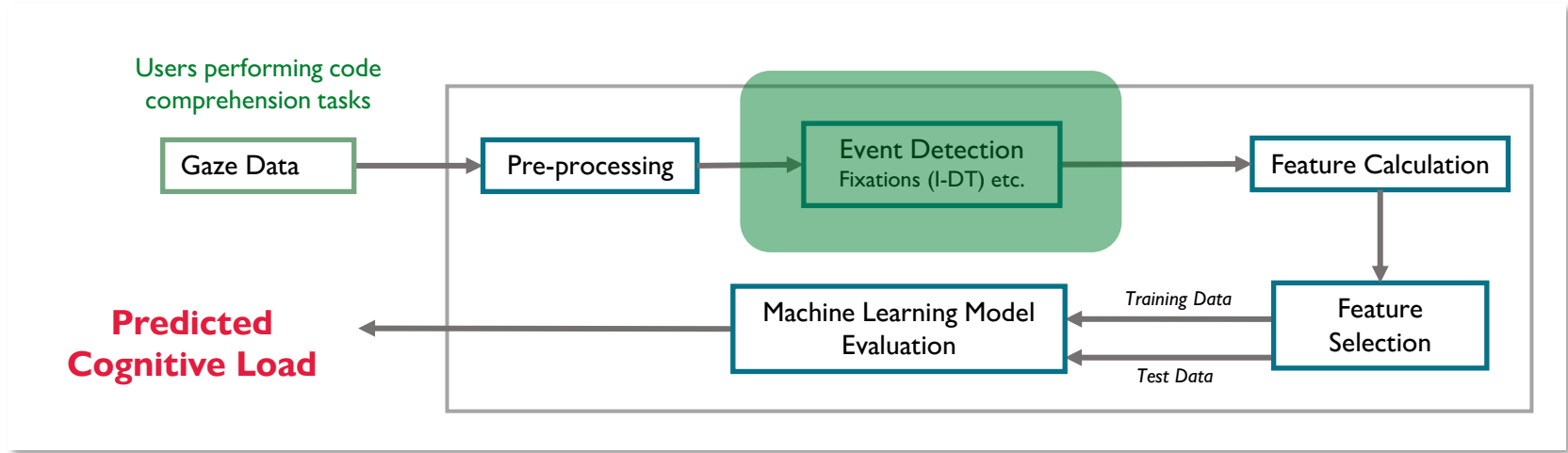
# Back to Our Pipeline



# Pre-processing

- **Pupil substitution:** Basic interpolation filter for pupil sizes.
- **Gazepoint substitution:** Basic interpolation filter for gaze points.
- **Blink detection:** Detect and remove blink artifacts.
- **Standard deviation:** Remove outliers identified by statistics.
- **Linear interpolation:** Linearly interpolate missing values.
- **Butterworth Filter:** Applies a third order lowpass Butterworth filter.

# Back to Our Pipeline



# Event Detection

Using the Velocity-Threshold Identification (I-VT) fixation classification algorithm

**Interpolation:** ☒

Max gap length in ms:

75

**Noise reduction:** ☐

Method:

Moving median

Window size in samples:

3

**Fixation filter:**

Window length in ms:

20

Velocity threshold in  
degrees/second:

30

**Discard short fixations:** ☒

Minimum fixation duration in  
ms:

60

**Merge adjacent fixations:** ☒

Maximum time between  
fixations in ms:

75

Maximum angle between  
fixations in degree:

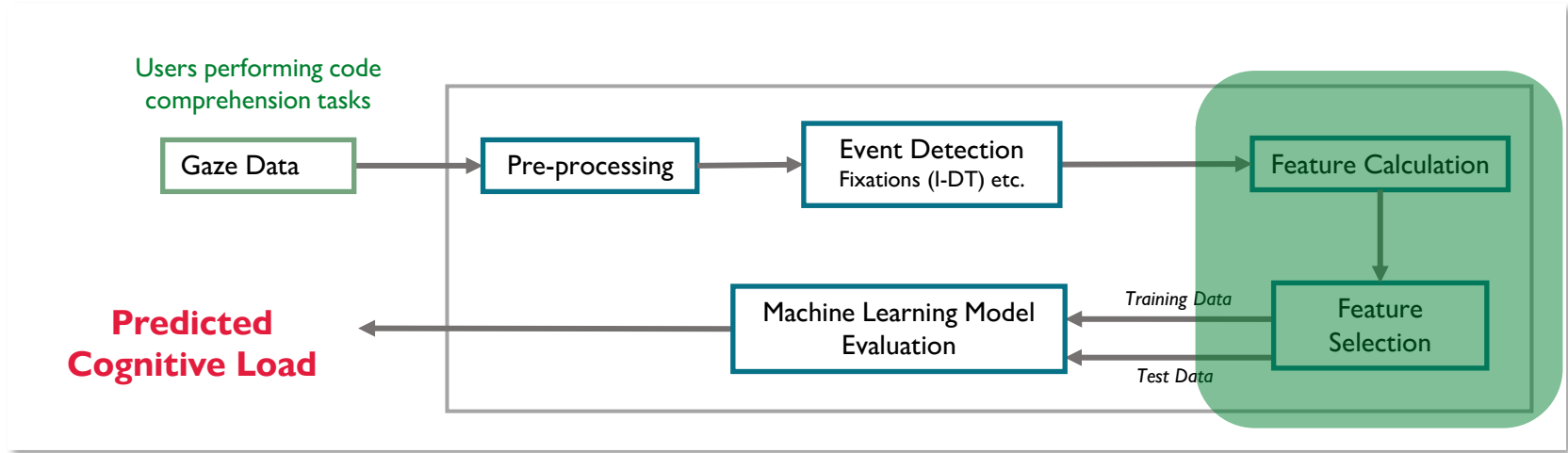
0.5

**Mapping Fixations to elements:**

Handling of fixation spanning  
over multiple elements, tabs  
or questions

Assign to the element/tab/question with max gaze points

# Back to Our Pipeline



# Feature Calculation and Selection

| Category              | Feature                                    | Description                                                                                                                                                                                                |
|-----------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fixation              | Fixation duration                          | Duration of the fixation in milliseconds                                                                                                                                                                   |
|                       | Fixation duration <i>fix.</i> $\geq 250ms$ | <i>Expression:</i> fixation duration <b>if</b> fixation duration $\geq 250ms$ <b>else</b> NaN                                                                                                              |
|                       | Fixation duration <i>fix.</i> $\geq 500ms$ | <i>Expression:</i> fixation duration <b>if</b> fixation duration $\geq 500ms$ <b>else</b> NaN                                                                                                              |
| Saccade               | Saccadic amplitude                         | Euclidean distance to the previous fixation in the (code) line and column coordinates                                                                                                                      |
|                       | No saccade                                 | <i>Expression:</i> 1 <b>if</b> source (i.e., previous) fixation is on the same line and column <b>else</b> 0                                                                                               |
|                       | Horizontal                                 | <i>Expression:</i> 1 <b>if</b> source fixation is on the same line and different column <b>else</b> 0                                                                                                      |
|                       | Non-horizontal                             | <i>Expression:</i> 1 <b>if</b> source fixation is on a different line <b>else</b> 0                                                                                                                        |
| Pupil                 | Pupil size <i>{aggr.}</i>                  | Pupil size in mm collected over the specified time window and aggregated using <i>{count, sum, max, min, mean, median, std}</i>                                                                            |
|                       | Number of pupil peaks                      | Number of pupil peaks observed in the specified time window                                                                                                                                                |
|                       | Amplitude of pupil peaks <i>{aggr.}</i>    | Amplitude of the individual pupil peaks observed in the specified time window and aggregated using <i>{count, sum, max, min, mean, median, std}</i>                                                        |
| (sub)<br>Scan-path    | Sub-scan length – # transitions            | Number of transitions between lines of code in the sub-scan derived over the specified time window. In our graph-based representation, this corresponds to the number of edges multiplied by their weights |
|                       | Sub-scan length – # unique visits          | Number of unique lines of code visited in the sub-scan derived over the specified time window. In our graph-based representation, this corresponds to the number of nodes in the graph                     |
|                       | Sub-scan density                           | (# transitions / # unique line visits)                                                                                                                                                                     |
|                       | Sub-scan entropy                           | Entropy of the sub-scan transition matrix. In our graph-based representation, this corresponds to the entropy of the graph adjacency matrix                                                                |
| Cluster-based<br>AOIs | Is fix. in spatial cluster                 | <i>Expression:</i> 1 <b>if</b> fixation is in spatial cluster <b>else</b> 0                                                                                                                                |
|                       | Is fix. in spatial-temporal cluster        | <i>Expression:</i> 1 <b>if</b> fixation is in spatial-temporal cluster <b>else</b> 0                                                                                                                       |

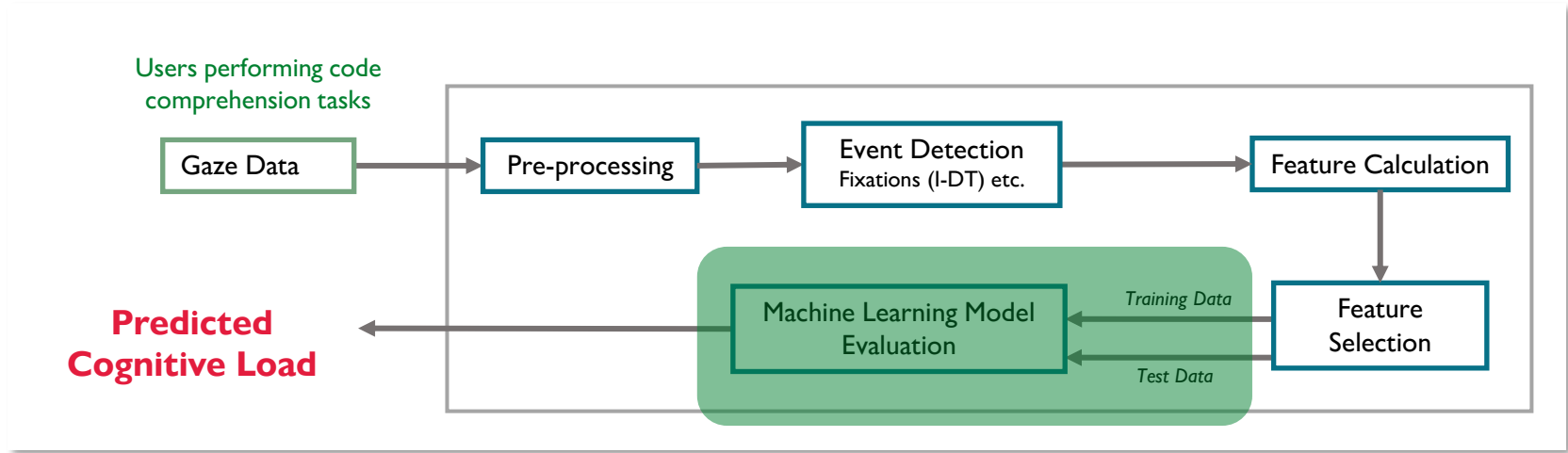
Features extracted at the fixation level. Abbreviations: Fix.: Fixation

# Feature Calculation and Selection

| Category      | Feature                                 | Description                                                        |
|---------------|-----------------------------------------|--------------------------------------------------------------------|
| Fixation      | Fixation count                          | Number of fixations on the line of code                            |
|               | Fixation count <i>fix.</i> $\geq 250ms$ | Number of fixations with duration $\geq 250ms$ on the line of code |
|               | Fixation count <i>fix.</i> $\geq 500ms$ | Number of fixations with duration $\geq 500ms$ on the line of code |
| Cluster-based | # Spatial clusters                      | Number of spatial clusters formed on the line of code              |
| AOIs          | # Spatial-temporal clusters             | Number of spatial-temporal clusters formed on the line of code     |

Additional features extracted at the line level. Abbreviations: Fix.: Fixation

# Back to Our Pipeline

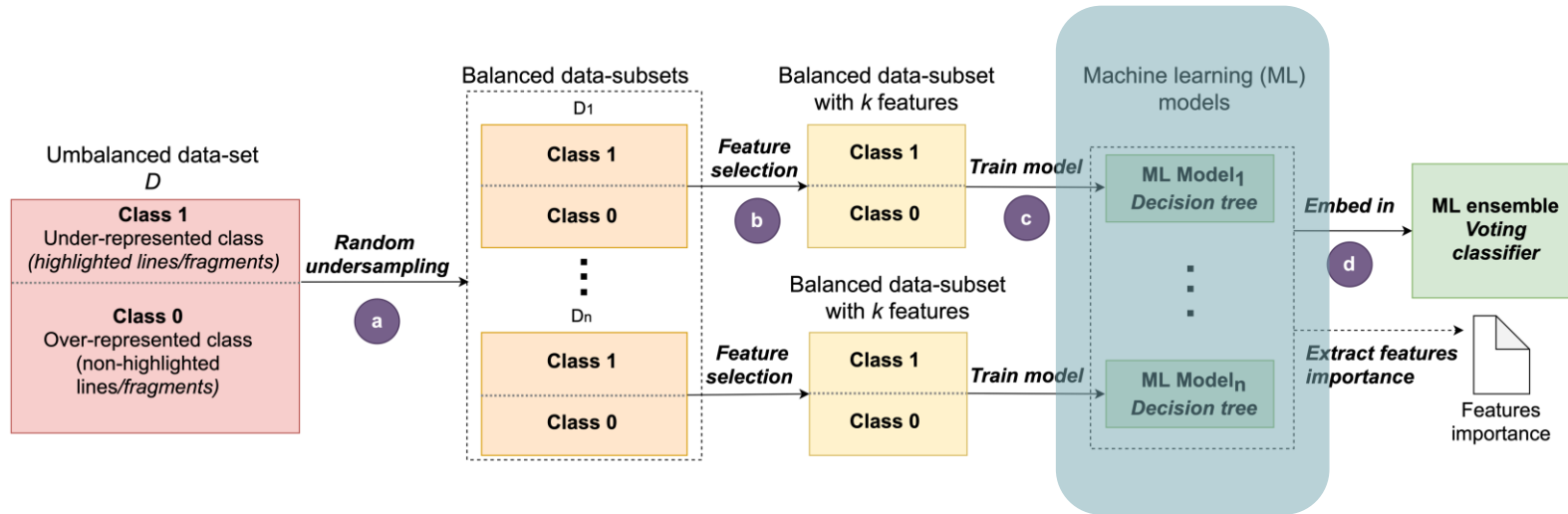




# Machine Learning Model Training

## Configuration

- **Supervised:** Ensemble learning
- **Features:** Cognitive load measures collected at a fine-grained level
- **Labels:** Participants' highlights of the mentally demanding parts of code



Architecture of the ensemble learning model used

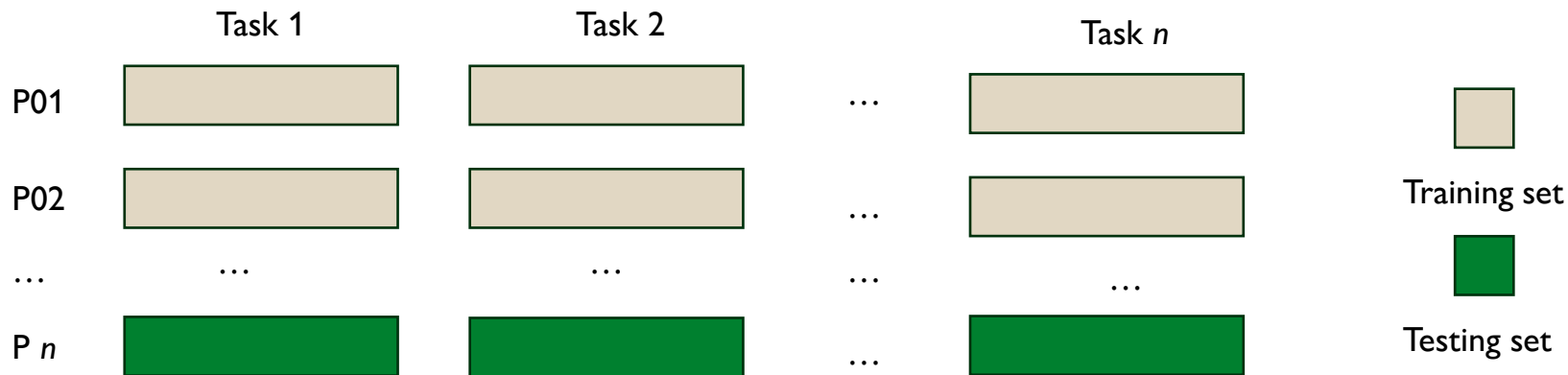
# Machine Learning Model Evaluation

## Cross-validation Variants

### Leave one participant out

Leave one task out

Leave one participant and task out



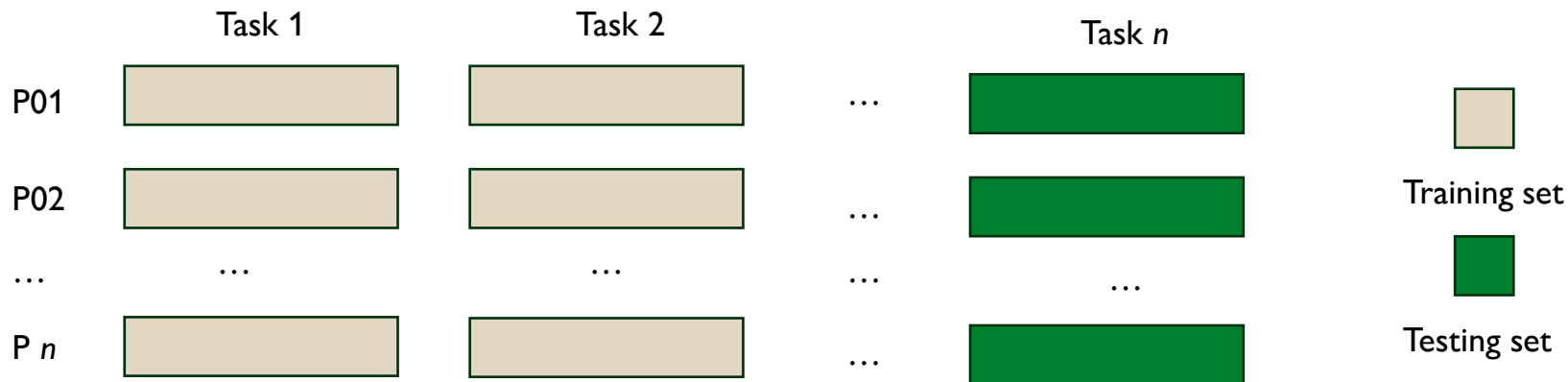
# Machine Learning Model Evaluation

## Cross-validation Variants

Leave one participant out

**Leave one task out**

Leave one participant and task out



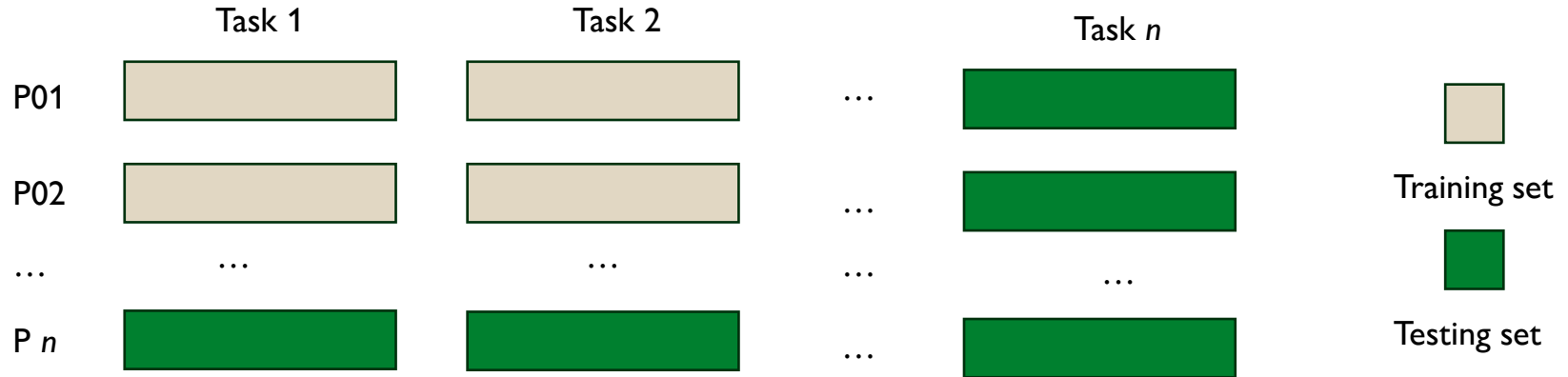
# Machine Learning Model Evaluation

## Cross-validation Variants

Leave one participant out

Leave one task out

**Leave one participant and task out**



# Machine Learning Model Evaluation

## Common Metrics

- **Accuracy:** Proportion of all predictions that are correct.  $(TP + TN) / Total$
- **Precision:** Of all predicted positives, how many are actually positive.  $TP / (TP + FP)$
- **Recall:** Of all actual positives, how many were correctly identified.  $TP / (TP + FN)$
- **F1 Score:** Harmonic mean of Precision and Recall; balances both into a single metric.  $2 \times (Precision \times Recall) / (Precision + Recall)$

## Example of Prediction results following the cross-validation approach

| Level    | Highlights | One-participant-out |        |        |        | One-task-out |        |        |        | One-participant-task-out |        |        |        |
|----------|------------|---------------------|--------|--------|--------|--------------|--------|--------|--------|--------------------------|--------|--------|--------|
|          |            | F1                  | Recall | Acc.   | Prec.  | F1           | Recall | Acc.   | Prec.  | F1                       | Recall | Acc.   | Prec.  |
| Line     | Initial    | 67.51%              | 67.36% | 69.94% | 71.23% | 65.17%       | 65.98% | 67.92% | 68.94% | 64.74%                   | 70.73% | 70.80% | 69.51% |
| Fragment | Initial    | 84.98%              | 85.47% | 85.16% | 86.25% | 80.79%       | 80.86% | 84.21% | 85.29% | 79.78%                   | 82.84% | 84.16% | 81.15% |
| Line     | Revised    | 68.57%              | 69.01% | 70.75% | 71.78% | 67.22%       | 67.43% | 70.30% | 71.60% | 65.96%                   | 73.35% | 71.48% | 68.96% |
| Fragment | Revised    | 85.65%              | 84.25% | 86.24% | 88.61% | 81.46%       | 80.26% | 84.47% | 85.82% | 81.14%                   | 84.32% | 85.56% | 82.21% |

# Machine Learning Model Evaluation

**Feature Importance:** A score indicating how much each input feature contributes to a model's predictions.

Higher importance means the feature has a stronger influence on the output.

## Example of feature importance results

| Feature gr.       | Ac. | Level | Highl.  | Feature gr. importance |              |              |
|-------------------|-----|-------|---------|------------------------|--------------|--------------|
|                   |     |       |         | OPO                    | OTO          | OPTO         |
| Fixation          | LI  | Line  | Initial | 0.102                  | 0.126        | 0.132        |
|                   | FI  | Frag. | Initial | 0.118                  | 0.093        | 0.11         |
|                   | LR  | Line  | Revised | 0.155                  | 0.162        | 0.167        |
|                   | FR  | Frag. | Revised | 0.082                  | 0.094        | 0.092        |
| Saccade           | LI  | Line  | Initial | 0.134                  | 0.151        | 0.157        |
|                   | FI  | Frag. | Initial | <b>0.653</b>           | <b>0.68</b>  | <b>0.655</b> |
|                   | LR  | Line  | Revised | 0.126                  | 0.135        | 0.115        |
|                   | FR  | Frag. | Revised | <b>0.702</b>           | <b>0.7</b>   | <b>0.694</b> |
| Pupil             | LI  | Line  | Initial | <b>0.587</b>           | <b>0.524</b> | <b>0.547</b> |
|                   | FI  | Frag. | Initial | 0.147                  | 0.159        | 0.164        |
|                   | LR  | Line  | Revised | <b>0.539</b>           | <b>0.563</b> | <b>0.553</b> |
|                   | FR  | Frag. | Revised | 0.151                  | 0.141        | 0.147        |
| Scan-path         | LI  | Line  | Initial | 0.174                  | 0.198        | 0.158        |
|                   | FI  | Frag. | Initial | 0.076                  | 0.062        | 0.064        |
|                   | LR  | Line  | Revised | 0.176                  | 0.135        | 0.158        |
|                   | FR  | Frag. | Revised | 0.059                  | 0.055        | 0.062        |
| Cluster-based AOI | LI  | Line  | Initial | 0.003                  | 0.002        | 0.006        |
|                   | FI  | Frag. | Initial | 0.006                  | 0.007        | 0.007        |
|                   | LR  | Line  | Revised | 0.004                  | 0.004        | 0.007        |
|                   | FR  | Frag. | Revised | 0.005                  | 0.01         | 0.006        |

# Offline Gaze Data Analysis for Activity Recognition

**We provide you with**

**Gaze dataset (*for the activity detection usecase*)**

- 20 participants
- 3 activities (Read, Inspect, Search)

**Jupyter Notebooks**

- Process and visualize gaze data
- Gaze Feature calculation
- SVM-based activity classifier

**You can**

**Explore these notebooks**

**And replace SVM  
with another model**

# Activity Prediction from Eye Movements

## 1. Data Collection

- Understand the activity: For example, *Reading* material (paper or digital etc.), geometric constraints, static vs. dynamic scene (scrolling) etc.
- Understand the temporal aspects that are relevant for data collection and model development (seconds to hours)
- Collect ground truth data

## 2. Model Development

- Calculate eye movement metrics (fixation, saccade, blinks, etc.)
- Data Preprocessing (cleaning, smoothing, interpolation, etc.)
- Calculate features (normalization)
- Split data to training and test sets
- Train models and document their performance

## 3. Feedback

- What do we do with the detected activity?

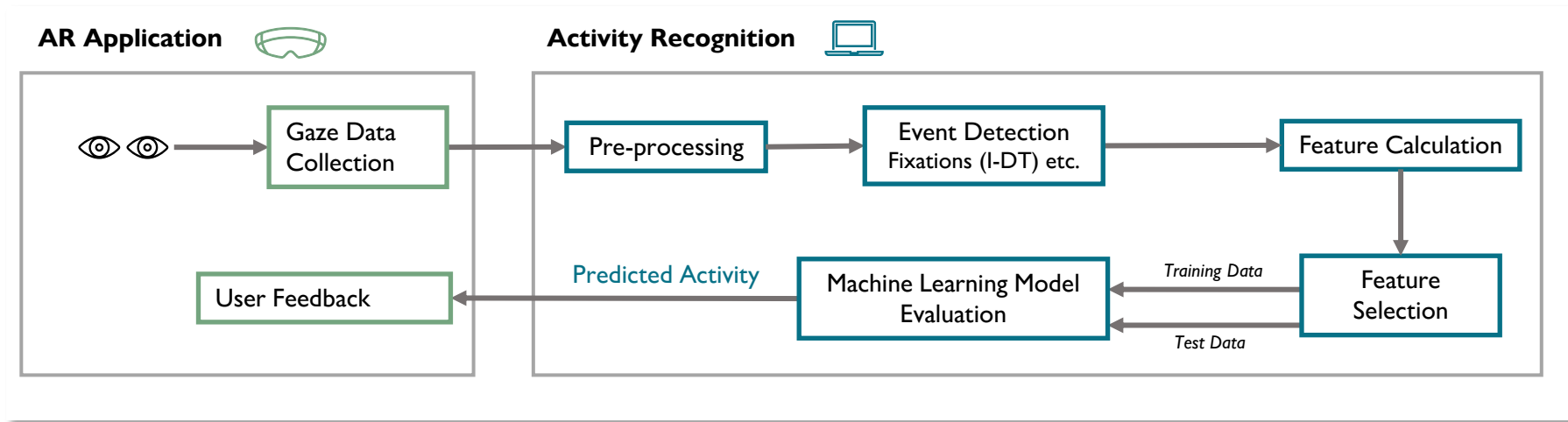


## Task 2 - Augmented Reality Activity Feedback (example)



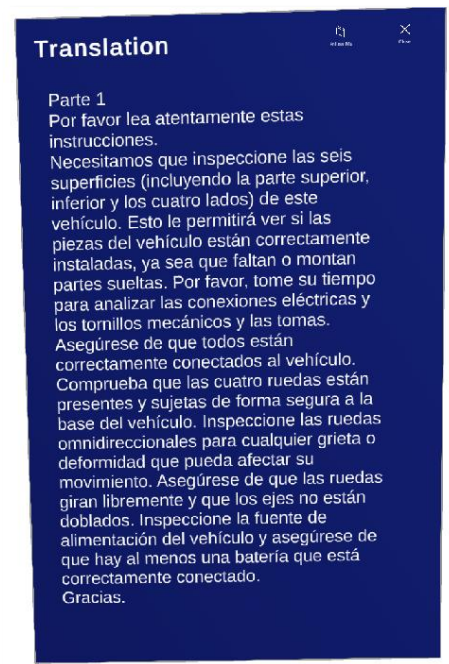
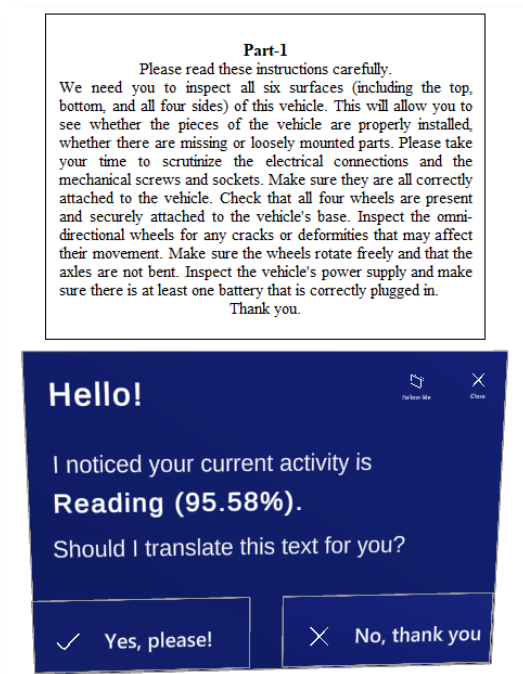
You will simulate a working online LLM-based feedback to the user that runs on the notebook. The feedback must be relevant to the current activity and provide some service linked with that activity.

### Pipeline:



## Task 2 - Augmented Reality Activity Feedback (example)

For the **Reading** activity, e.g., translating the text.



# GEAR Video - Reading

## 1st Activity: Reading

Link to video: <https://www.youtube.com/watch?v=Dq-Z5p61J8E>

# GEAR Video - Inspection

## 2nd Activity: Inspection

Link to video: <https://www.youtube.com/watch?v=Dq-Z5p61J8E>

# GEAR Video - Search

## 3rd Activity: Search

Link to video: <https://www.youtube.com/watch?v=Dq-Z5p61J8E>

## Notebook 4:



|                                           |                                         |
|-------------------------------------------|-----------------------------------------|
| 📁 Data/RawGazeData                        | Delete Data/RawGazeData/placeholder.txt |
| 📁 demo_images                             | intervention example added              |
| 📄 AnSVMClassifierForHL2GazeFeatures.ipynb | Add files via upload                    |
| 📄 FeatureCalculation.ipynb                | Add files via upload                    |
| 📄 InterventionExample.ipynb               | intervention example added              |
| 📄 OnlineFeaturesCalculation.ipynb         | Add files via upload                    |
| 📄 README.md                               | Initial commit                          |
| 📄 ReadAndVisualizeRawHL2GazeData.ipynb    | Add files via upload                    |

# Experimenting with Different LLMs

```
response = client.chat.completions.create(  
    model="openai/gpt-4o",  
    messages=[  
        {  
            "role": "user",  
            "content": [  
                {"type": "image_url", "image_url": {"url": f"data:image/png;  
base64,{b64_image}"},  
                {"type": "text", "text": PROMPTS[detected_task]},  
            ],  
        },  
    ],  
)
```

InterventionExample.ipynb

You need to choose a multi-modal LLM that accepts both text and images

Check for available models in OpenRouter and use the API key that we will provide to you

| MODALITIES | INPUT PRICE | OUTPUT PRICE | CONTEXT | WEEKLY TOKENS |
|------------|-------------|--------------|---------|---------------|
|            | \$5 per 1M  | \$30 per 1M  | 1M      | 491B          |

Pay attention to the allowed modalities and cost

# Task 2 – LLM-based User Feedback

**We provide you with**

**A Notebook that can take**

**An activity**

**An image of the current scene**

**An example prompt**

**To generate a relevant feedback**

**You can**

**Explore this notebook**

**And replace the input image and  
existing prompt to explore the  
potential of the current pipeline**



# Interactive Session

## In groups

- Think of a **domain** where developing an adaptive system is relevant (e.g., e-learning, software development, healthcare or any domain possibly related to your research or job activities)
- Discuss the following **value questions** about your adaptive system
  - **Who** are the users of your adaptive system?
  - **What** is the purpose of the adaptive system?
  - **When** are the adaptive system and associated intervention mechanisms used?
  - **Where** is the adaptive system accessed?
  - **Why** are the adaptive system and associated intervention mechanisms needed?
  - **How** do the adaptive system and associated intervention mechanisms deliver value?

# Interactive Session

## In groups

- Discuss the following technical questions about your adaptive system
  - What data (features and label) do you need for training the detection model (e.g., to predict user activity or cognitive load)
  - How would the data be collected, in what environments and settings?
  - How will you train the model and what evaluation approach would you use?
  - How would you leverage LLMs to generate interventions?
    - Generate a few example prompts, test them with an LLM using the provided notebook
    - Reflect on the quality of the responses obtained and any opportunities or challenges perceived with the use of your prompts
  - How would you evaluate the effectiveness of your interventions?

# Interactive Session

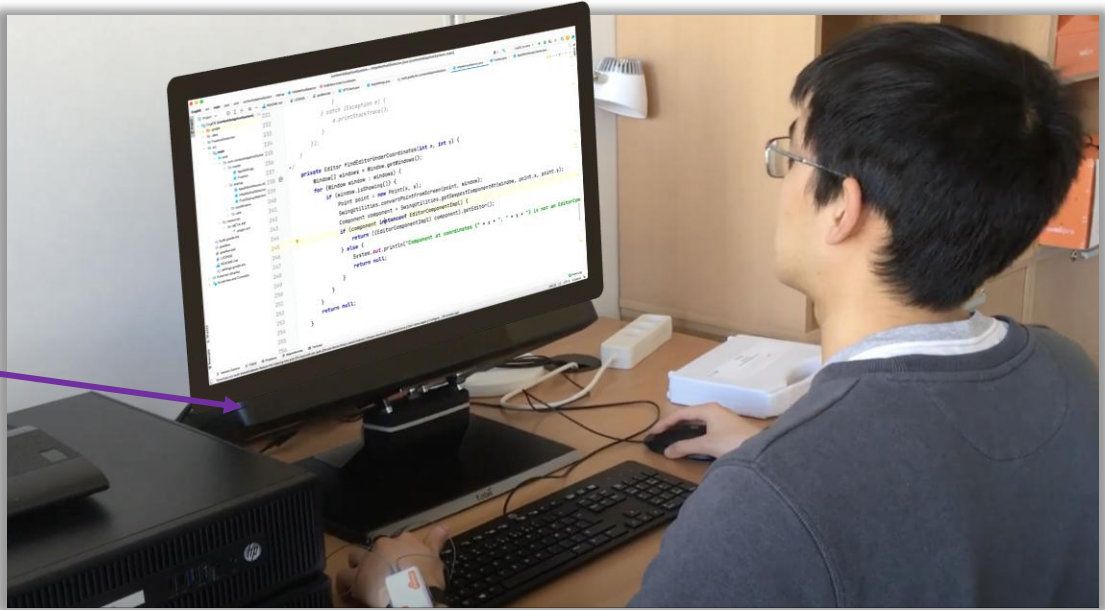
You can draw a flow chart or create a storyboard to describe the control flow of your system and associated intervention mechanisms

- Prepare a 3 to 5-minute pitch to present your adaptive system and associated intervention mechanisms

## Example Idea:

*While working on a large code base (e.g., Java project), automatically detect the parts of the source code that are difficult to understand and provide the user contextually relevant feedback*

A screen-based  
Eye-tracker



# Example Idea

The screenshot shows an IDE window titled "contextAdaptiveSystem - HttpMethodSelector.java [contextAdaptiveSystem.main]". The code is a Java file with a method `findEditorUnderCoordinates`. A red dashed box highlights a section of the code and an explanatory text block on the right.

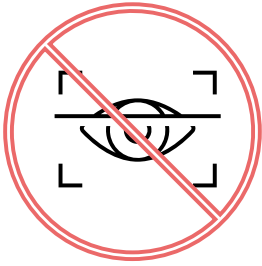
```
231 }
232 } catch (Exception e) {
233     e.printStackTrace();
234 }
235 }
236 }
237 */
238 @ private Editor findEditorUnderCoordinates(int x, int y) {
239     Window[] windows = Window.getWindows();
240     for (Window window : windows) {
241         if (window.isShowing()) {
242             Point point = new Point(x, y);
243             SwingUtilities.convertPointFromScreen(point, window);
244             Component component = SwingUtilities.getDeepestComponentAt(
245                 window, point.x, point.y);
246             if (component instanceof EditorComponentImpl) {
247                 return ((EditorComponentImpl) component).getEditor();
248             } else {
249                 System.out.println("Component at coordinates (" + x + ", " + y + ") is not an EditorComponentImpl");
250                 return null;
251             }
252         }
253     }
254     return null;
255 }
```

High cognitive load was detected when reading `findEditorUnderCoordinates()`!

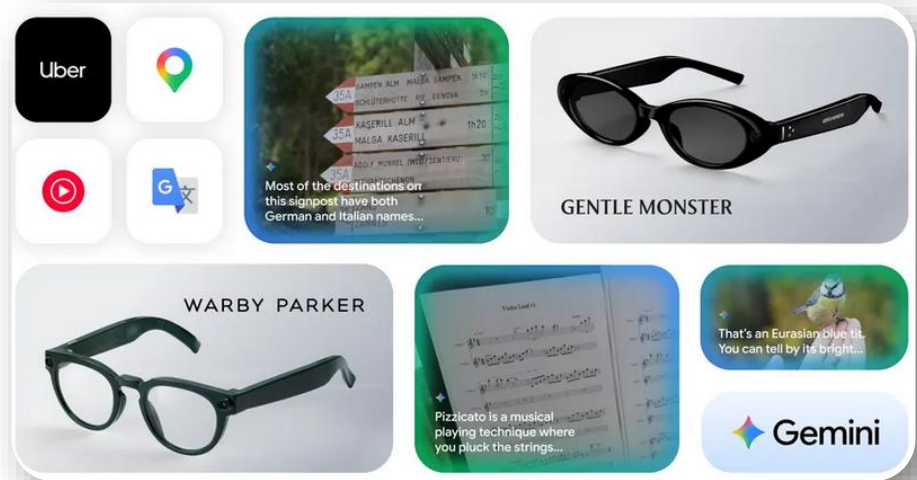
The following is a brief explanation of this method:

- **Retrieve Windows:** The method gathers all windows currently associated with the application.
- **Filter Visible Windows:** It then checks each window to see if it is currently visible on the screen.
- **Coordinate Conversion:** The screen coordinates provided are converted into coordinates relative to each window being checked.
- **Find Deepest Component:** Within each visible window, the method finds the most deeply nested component at the specified coordinates.
- **Check Component Type:** It checks if this component is of a specific type (`EditorComponentImpl`).
- **Return Editor:** If the component is the expected type, the method retrieves and returns the associated `Editor` object.
- **Handle Non-Matching Component:** If the component is not the expected type, it logs a message and returns `null`.

Add comments to Refactor the code



## Google's Gentle Monster and Warby Parker



# Thank You!

## Dr. Kenan Bektaş

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programming-and-development/](https://ics.unisg.ch/chairs/barbara-weber-software-systems-programming-and-development/)

